

RAJALAKSHMI ENGINEERING COLLEGE

RAJALAKSHMI NAGAR, THANDALAM – 602 105



RAJALAKSHMI ENGINEERING COLLEGE

CS23A34

USER INTERFACE AND DESIGN LAB

Laboratory Observation NoteBook

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Semester : IV

Academic Year: 2024-25

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Create a data visualization (e.g., pie charts, bar graphs) for an inventory management system using javascript

AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

Step 1: Set Up Your HTML File

First, create an HTML file to hold your canvas for the chart and include Chart.js.

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8" />
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0"/>
```

```
<title>Inventory Management Visualization</title>
```

```
<style>  body {    font-family:
Arial, sans-serif;    text-align:
center;                margin: 50px;
background-color: #f5f7fa;

    }
```

```
    h1 {    color: #2c3e50;
margin-bottom: 40px;

    }
```

```
    .charts-container    {
display: flex;            flex-
wrap: wrap;            justify-
content: center;        gap:
40px;

    }
```

```
    canvas {            background-color: #fff;
border-radius: 8px;        box-shadow: 0 5px
15px rgba(0,0,0,0.1);    padding: 10px;
```

```

    }

</style>

</head>

<body>

    <h1>Inventory Management System</h1>

    <div class="charts-container">

        <canvas id="pieChart" width="400" height="400"></canvas>

        <canvas id="barChart" width="400" height="400"></canvas>

    </div>

    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

    <script src="script.js"></script>

</body>

</html>

```

Step 2: Create the JavaScript File for Charts

Next, create a JavaScript file (script.js) to handle the data visualization logic.

```

// Data for the inventory const
inventoryData = {
  labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'], datasets:
  [
    {

```

```

    label: 'Items in Stock',
    data: [200, 150, 100, 80, 50],
    backgroundColor: [
        '#FF6384',
        '#36A2EB',
        '#FFCE56',
        '#4BC0C0',
        '#9966FF'
    ]
  }
]
};

```

// Creating the Pie Chart

```

const ctxPie = document.getElementById('pieChart').getContext('2d'); const
pieChart = new Chart(ctxPie, {
  type: 'pie', data:
inventoryData,
options: {
responsive: true,
  plugins: {
title: {
display: true,
  text: 'Inventory Distribution'
  }
  }
  }
});

```

// Creating the Bar Chart

```

const ctxBar = document.getElementById('barChart').getContext('2d'); const
barChart = new Chart(ctxBar, {
  type: 'bar', data:
inventoryData,
options: {
responsive: true,

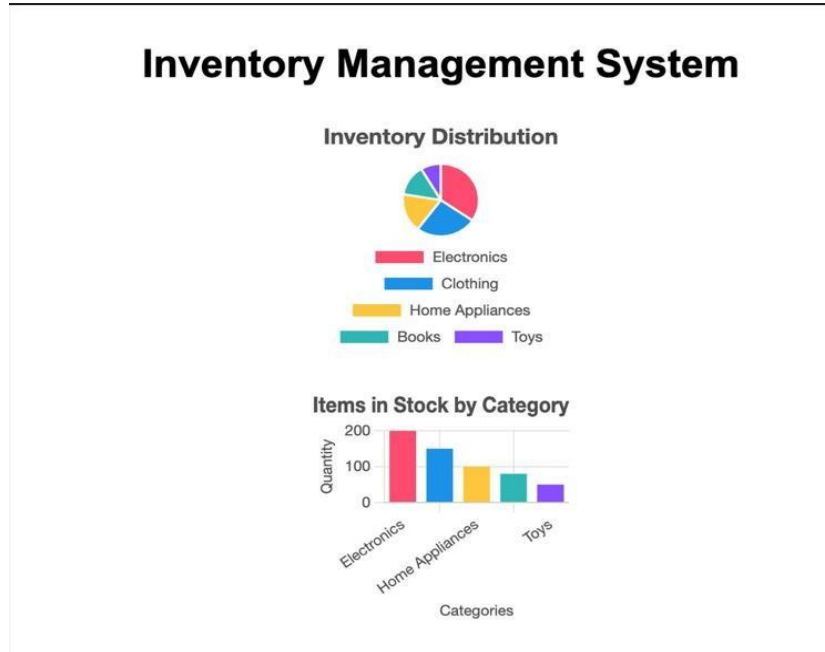
```

```

plugins: {      title: {
display: true,
  text: 'Items in Stock by Category'
} },
scales: {
  y: {
    beginAtZero: true
  }
}
});

```

OUTPUT:



RESULT:

Data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript has been successfully created.